David Aretz

Max Planck Institute for Mathematics
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Research interests

Higher differential geometry; supersymmetric functorial field theories; homotopy theory and K-theory; noncommutative geometry.

Education

2023–present
Doctoral studies in mathematics
University of Bonn and Max Planck Institute for Mathematics, Bonn.
Supervisors: Christian Blohmann and Peter Teichner.
Expected completion: September 2027.
Provisional thesis title: *Classifying spaces of supersymmetric field theories*.

2020–2023
M.Sc. Mathematics, University of Bonn
Final grade: 1.0, with distinction. Supervisor: Christian Blohmann.
Thesis: *Cohomological Extension of Groupoid Convolution*.
davidaretz.github.io

2017–2020
B.Sc. Mathematics, University of Bonn
Final grade: 1.0, with distinction. Secondary subject: Astrophysics.
Supervisor: Christian Blohmann.
Thesis: *The Noncommutative Geometry of Symplectic Singularities*.
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Publications & Preprints

2026
Functoriality of bornological groupoid convolution
David Aretz and Christian Blohmann.
Journal of Functional Analysis **291**, no. 12, article 111618.
doi:10.1016/j.jfa.2026.111618 · arXiv:2508.11005

2026
Super K-theory and group completion
David Aretz and Luuk Stehouwer.
Preprint, submitted. 98 pp.
arXiv:2609.02407

2026
Morita equivalence and the $\mathbb{E}_\infty$-spin orientation
David Aretz.
Preprint.

2026
Elliptic objects and supersymmetric field theories
David Aretz and Daniel Berwick-Evans.
Lecture notes, mini-course at the Bernoulli Center, Lausanne, July 2026. 33 pp.

Work in progress

In preparation
Twisted supersymmetric field theories in dimension zero
David Aretz and Karolis Dembickas.
Expected late 2026–early 2027.

Research talks and conference presentations

November 2026 — **The higher algebra of the spin orientation.**
Algebra/Topology Seminar, Copenhagen.

October 2026 — **K-theory for super algebras**
Texas Tech Topology and Geometry Seminar.

July 2026 — **Super K-theory**
IMPRS Seminar, MPIM Bonn.

July 2026 — **The higher algebra of the spin orientation**
Elliptic Objects, von Neumann Algebras and Functorial Field Theory, Lausanne.

May 2026 — **K-theory for super algebras**
Higher Differential Geometry, Greifswald.

October 2025 — **Functoriality of Lie groupoid convolution algebras**
Higher Differential Geometry Seminar, MPIM Bonn.

July 2025 — **Functoriality of Lie groupoid convolution algebras**
Invited talk, Workshop on Singular Foliations and Higher Structures, Würzburg.

June 2025 — **The algebra of smooth functions on a differentiable stack**
Invited talk, Higher Structures Seminar, Göttingen.

March 2025 — **Categorical perspectives on Clifford algebras and geometric spin structures**
BCGW PhD Retreat, Burbach.

May 2024 — **Homotopical vector bundles on differentiable stacks**
Gong talk, From Analysis to Homotopy Theory, Greifswald.

January 2024 — **Equivariant vector bundles in supergeometry**
WIQI Topology Seminar, MPIM Bonn.

October 2023 — **Odd parallel transport, twists and local systems**
BGW PhD Retreat.

Teaching experience

Winter 2026/27 — **Higher Category Theory**
Tutor, with Merlin Christ.

2022, 2024, 2025, 2026 — **Workshops for new tutors**
University of Bonn, with Regula Krapf.
Leading one of the one-day workshops for new tutors in the Mathematical Institute.

2018–2023 — **Tutor, Mathematical Institute, University of Bonn**
September 2018–March 2023.

Geometry and Analysis on Manifolds	Winter 2021/22, 2022/23
Riemannian Geometry	Summer 2022
Analysis II	Summer 2021
Analysis I	Winter 2020/21
Complex Analysis	Summer 2020
Analysis III and revision course	Winter 2019/20
Mathematics precourses	2018, 2019

Academic service and organization

2026–present **Co-organizer, Bonn Math & AI PhD events**
With Fabio Neugebauer and Emma Pechouckova.

2026–present **Co-organizer, Quantum Topology Seminar**
With Aaron Hofer, Naageswaran Manikandan, Léo Schelstraete and Filippos Sytilidis.

2024–2025 **Co-organizer, Physical Mathematics Seminar**
With David Prinz, Max Planck Institute for Mathematics.

March 2025 **Leading organizer, BCGUW PhD Retreat**
Higher Structures in Differential Geometry, Burbach.
Bonn–Cologne–Göttingen–Utrecht–Würzburg.

October 2024 **Leading organizer, BGW PhD Retreat**
Higher Structures in Differential Geometry, Bielefeld.
Bonn–Göttingen–Würzburg.

Awards and scholarships

2020–2023 **Scholarship, Studienstiftung des deutschen Volkes**

2020 **Prize for the best bachelor's degree in Mathematics**
Bonner Mathematische Gesellschaft.

2019–2020 **Honors Program, University of Bonn**
January 2019–August 2020.

Parental leave

June–July 2024 **Parental leave**